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Exactly mergeable summaries

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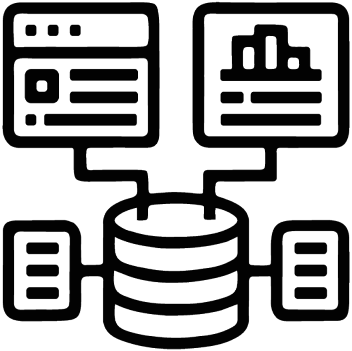

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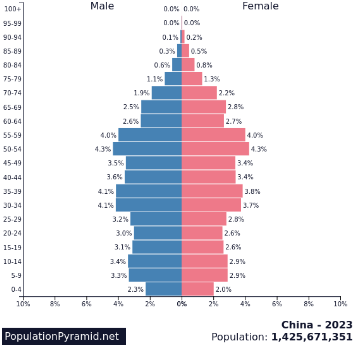
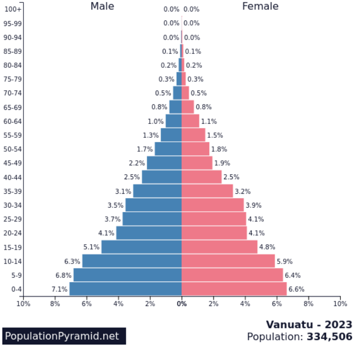
Motivation





$$(n_{A\cup B}, \mathbf{p}_{A\cup B}) = (n_A + n_B, \frac{n_A\mathbf{p}_A + n_B\mathbf{p}_B}{n_A + n_B})$$

Aggregation

Data analysis programs provide aggregation functions such as means (arit, geom, harm, median, modus), min, max, product, bounded sum, counting, etc.

A problem with traditional aggregation is that often too much information is discarded thus reducing the precision of the obtained results.

A much better, preserving more information, summarization of original data can be achieved by representing aggregated data using selected types of complex data such as symbolic objects (Diday, 1988, 1995), compositions (Aitchison, 1986), functional data (Ramsay and Silverman, 2005), etc.

Mergeable summaries

In complex data analysis the measured values over a selected subset A of the set of units U are aggregated into a complex object $\Sigma(A)$ and not into a single value. Most of the standard aggregation theory does not apply directly.

In our contribution, we present an attempt to start building a theoretical background of complex aggregation.

An interesting question is, which complex data types are compatible with the merging of disjoint subsets of units

$$\Sigma(A \cup B) = F(\Sigma(A), \Sigma(B)), \qquad \text{for} \quad A \cap B = \emptyset. \tag{1}$$

Mergeable summaries were proposed and elaborated by Agarwal et al. (2012). They enable parallelization in big data algorithms and stream processing.

Exactly mergeable summaries

The summarization in big data is not deterministic and allows some errors. A summary is *mergeable* if the error and space (size of summary) do not increase after the merge.

We will discuss exactly mergeable summaries "without errors".

A summary $\Sigma(A)$ is an *exactly mergeable summary* if and only if it requires a fixed space of small size and satisfies the relation (1).

We assume that a numerical variable v is measured on the set of units U and that $A, B \subseteq U$ and $A \cap B = \emptyset$.

Let $\text{sort}_A(v)$ be a list of values of the variable v ordered in decreasing order on the subset of units A. We define $\text{1st}_A(v) = \text{sort}_A(v)[1]$ and $\text{2nd}_A(v) = \text{sort}_A(v)[2]$.

Exactly mergeable summaries

simple examples

1. $\Sigma(A) = |A| = n_A$
   $\Sigma(A \cup B) = \Sigma(A) + \Sigma(B)$
2. $\Sigma(A) = \min_{X \in A} v(X)$
   $\Sigma(A \cup B) = \min(\Sigma(A), \Sigma(B))$
3. $\Sigma(A) = \max_{X \in A} v(X)$
   $\Sigma(A \cup B) = \max(\Sigma(A), \Sigma(B))$
4. $\Sigma(A) = (\mathrm{1st}_A(v), \mathrm{2nd}_A(v))$
   $\Sigma(A \cup B) = (\mathrm{1st}_L(v), \mathrm{2nd}_L(v))$,
   where $L = \{\mathrm{1st}_A(v), \mathrm{2nd}_A(v), \mathrm{1st}_B(v), \mathrm{2nd}_B(v)\}$
   This example can be generalized to $\Sigma(A) = \text{Top-}k_A(v)$.
5. $\Sigma(A) = (n_A, \mu_A), \quad \mu_A = \frac{1}{n_A} \sum_{X \in A} v(X)$
   $\Sigma(A \cup B) = (n_A + n_B, \frac{n_A \mu_A + n_B \mu_B}{n_A + n_B})$
6. $\Sigma(A) = \sum_{X \in A} v(X)$
   $\Sigma(A \cup B) = \Sigma(A) + \Sigma(B)$

Exactly mergeable summaries

intervals

The *interval summary* of the variable v on the set of units A is defined as

$$\Sigma(A) = [\min_{X \in A} v(X), \max_{X \in A} v(X)]$$

It is an exactly mergeable summary.

Proof: Let $\Sigma(A) = [m_A, M_A]$ and $\Sigma(B) = [m_B, M_B]$ then

$$\Sigma(A \cup B) = [\min_{X \in A \cup B} v(X), \max_{X \in A \cup B} v(X)] = [\min(m_A, m_B), \max(M_A, M_B)]$$

Exactly mergeable summaries

moments

The distribution of values of variable v on the set of units A is often summarized by its average μ_A and its standard deviation σ_A. It would be better to represent it as $\Sigma(A) = (n_A, \mu_A, \sigma_A)$, where n_A is the number of units in A.

Then the distribution of additional values of variable v on the set of units B, $A \cap B = \emptyset$, is summarized by $\Sigma(B) = (n_B, \mu_B, \sigma_B)$ and can be combined into a summary of the distribution on the set $C = A \cup B$, $\Sigma(C) = (n_C, \mu_C, \sigma_C)$ determined by $\Sigma(A)$ and $\Sigma(B)$ as follows

$$n_C = n_{A \cup B} = n_A + n_B$$

$$\mu_C = \mu_{A \cup B} = \frac{n_A \mu_A + n_B \mu_B}{n_C}$$

$$\sigma_C = \sigma_{A \cup B} = \sqrt{\frac{S_C}{n_C} - \mu_C^2}$$

where $S_C = S_A + S_B$ and $S_X = n_X(\sigma_X^2 + \mu_X^2)$. $\Sigma(A)$ is an exactly mergeable summary. Can be extended to higher moments.

Exactly mergeable summaries

set membership count

Counting the number of units from C in A

$$n(A; C) = |A \cap C|$$

is an exactly mergeable summary.

Proof:

$$n(A \cup B; C) = |(A \cup B) \cap C| = |(A \cap C) \cup (B \cap C)| =$$

$$= |A \cap C| + |B \cap C| - |A \cap B \cap C| = n(A; C) + n(B; C)$$

Combining exactly mergeable summaries

Let Σ_1 and Σ_2 be exactly mergeable summaries. Then also

$$\Sigma_1 \oplus \Sigma_2(A) = (\Sigma_1(A), \Sigma_2(A))$$

is an exactly mergeable summary.

Proof: $\Sigma_1 \oplus \Sigma_2(A \cup B) = (\Sigma_1(A \cup B), \Sigma_2(A \cup B)) =$

$$= (F_1(\Sigma_1(A), \Sigma_1(B)), F_2(\Sigma_2(A), \Sigma_2(B)))$$

Combining exactly mergeable summaries

Let $K = \{k_1, k_2, \ldots, k_s\}$ be a finite set of categories and $v : U \to K$ a categorical (nominal) variable on the set of units U. The summary

$$\Sigma(A) = \{(k, n(A, C(k))) : k \in K\} \quad \text{where} \quad C(k) = \{X : v(X) = k\}$$

is called a *bar chart*.

Let $v : U \to \mathbb{R}$ be an ordinal variable and $\mathbf{B} = (B_1, B_2, \ldots, B_r)$ an ordered partition (set of bins) of $v(A)$. The summary

$$\Sigma(A) = [(B, n(A, C(B))) : B \in \mathbf{B}] \quad \text{where} \quad C(B) = \{X : v(X) \in B\}$$

is called a *histogram*.

Therefore, since set membership counts are exactly mergeable, the bar charts and histograms are exactly mergeable summaries.

Proving that a summary is not exactly mergeable

If for a summary Σ exist sets A_1, B_1, A_2, B_2 such that $A_1 \cap B_1 = \emptyset$, $A_2 \cap B_2 = \emptyset$, $\Sigma(A_1) = \Sigma(A_2)$, $\Sigma(B_1) = \Sigma(B_2)$, and $\Sigma(A_1 \cup B_1) \neq \Sigma(A_2 \cup B_2)$ then Σ **is not** exactly mergeable.

Proof: Assume that Σ is exactly mergeable. Then

$$\Sigma(A_1 \cup B_1) = F(\Sigma(A_1), \Sigma(B_1)) = F(\Sigma(A_2), \Sigma(B_2)) = \Sigma(A_2 \cup B_2)$$

– a contradiction.

Median is not exactly mergeable summary

$$\mathrm{med}_A(v) = \mathrm{sort}_A(v)[\lceil \frac{n_A}{2} \rceil]$$

$$\begin{array}{ll} v(A_1) = [3, 4, 1] & \mathrm{med}_{A_1}(v) = 3 \\ v(B_1) = [9, 6] & \mathrm{med}_{B_1}(v) = 6 \\ & \mathrm{med}_{A_1 \cup B_1}(v) = 4 \end{array}$$

$$\begin{array}{ll} v(A_2) = [3, 8] & \mathrm{med}_{A_2}(v) = 3 \\ v(B_2) = [6, 2, 7] & \mathrm{med}_{B_2}(v) = 6 \\ & \mathrm{med}_{A_2 \cup B_2}(v) = 6 \end{array}$$

Second is not exactly mergeable summary

$$\mathrm{2nd}_A(v) = \mathrm{sort}_A(v)[2]$$

$$\begin{array}{ll} v(A_1) = [1,3,5] & \mathrm{2nd}_{A_1}(v) = 3 \\ v(B_1) = [2,5,6] & \mathrm{2nd}_{B_1}(v) = 5 \\ & \mathrm{2nd}_{A_1 \cup B_1}(v) = 2 \end{array}$$

$$\begin{array}{ll} v(A_2) = [3,3,6] & \mathrm{2nd}_{A_2}(v) = 3 \\ v(B_2) = [4,5,7] & \mathrm{2nd}_{B_2}(v) = 5 \\ & \mathrm{2nd}_{A_2 \cup B_2}(v) = 3 \end{array}$$

Conclusions

- We introduced the notion of exactly mergeable summaries. We showed that the summaries interval, max, min, top-k, bar chart, and histogram (used in SDA) are exactly mergeable.
- Extending a summary with the size of the set of units makes some summaries, such as moments and discrete probability distribution, exactly mergeable.
- Develop symbolic methods for exactly mergeable summaries.

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